

Davvey's Vehicle Rigging Help Files.



Index:-

1.Introduction

2.Installation

3.Initial Usage

4.Steering

5.Wheel Auto Rotate

6.Wheel spin

7.Bank control

8.Suspension

9.Tilt Control

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Introducton.

Welcome and thanks for Downloading these scripts.

This is a series of scripts to set up and rig a Vehicle using Utility nodes instead of any expressions.

The vehicle can be setup to "AutoRotate" the wheels by moving the vehicle in the X and Z planes. Rotation of the rig in the Y Plane will automatically set wheel rotation to compensate of vehicle moving in the opposite direction after rotation. Also additional spin can be added or blended on top to simulate wheel spin etc. Also each wheel can be switched on or off, also each wheel can be set to stop or rotate in reverse at any time by keying the relevant attributes.

Vehicle rig also sets up steering for front (and rear) with independent steering for both front and rear steer with different ratios for both if required. Also front and rear can be switched off or even reversed to simulate the type of steering seen on high performance cars.

The rig also sets up fully independent and controllable suspension for each wheel. Each Wheel travel can be limited if required. The vehicle will tilt and swivel based on position of wheel ground locators. If these are constrained to a ground plane then the suspension will automatically operate.

Main procedure uses a variable $GroupV_A to store main groups from GUI and build a rig of joints with locators as required.

Joints are constrained to individual locators. Also some locators are constrained to individual joints. I am currently working on building a full suite of scripts to allow rigging of all the various parts of any wheeled vehicle. The intention is to build a set of modular scripts to allow rigging of individual parts and or a fully rigged vehicle.

This script current has no error checking if you find it useful please let me know and I will rewrite it with full error checking.

Installation instructions.

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To install copy all scripts to your User Script directory (or any script directory on the Maya Script path environment).

The scripts are all called from the DVY_VehicleRigger GUI.

Also you can type names in the field as required. Script will error out if not all items are found. Once all fields correct press Rig Vehicle button.

These scripts currently have minimal error checking but if you find it useful please let me know and I will rewrite it with full error checking.

This script was written to demonstrate the use of Utility nodes to replace expressions for vehicle control etc. I am currently working on building a full suite ot scripts to allow rigging of all the various parts of any wheeled vehicle. The intention is to build a set of "modular" scripts to allow rigging of individual parts and or a fully rigged vehicle.

Directions:

Once installed restart Maya or "rehash" Maya to ensure all scripts are sourced. Source and run DVY_VehicleRigger.mel, use default values if you have named them as shown or select each item in turn and load required field. Also you can type names in the field as required. Script will error out if not all items are found. Once all fields correct press "Rig Vehicle" button.

Each Tab must be initialized by selecting the relevant CheckBox on the first Tab. To set up the Doors and Hood Section, select the required items using the check boxes, then select the items or use default values (same as wheel section method). Once all field completed press the "Set Hinge Position" Button on that Tab. Repeat for Hood etc Tab if required , then adjust position of the Blue Spheres to the exact position you want the Hinge point to be. Then adjust the Red Spheres to the`Handle` position for item item. The Blue Spheres can be Translated and Rotated, the Red just Translated. The Red handle sphere is used to set the angle of rotation of the Hinge. The limit of opening angle is 110 degrees but this can be changed using the Controllers `Y` Translate Axis limits in the Attribute Editor.

The Window Spheres (green and Yellow) carry out a similar function but are used to set up a sliding action to simulate Window opening.

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Davvey's Vehicle Rigging Help Files.



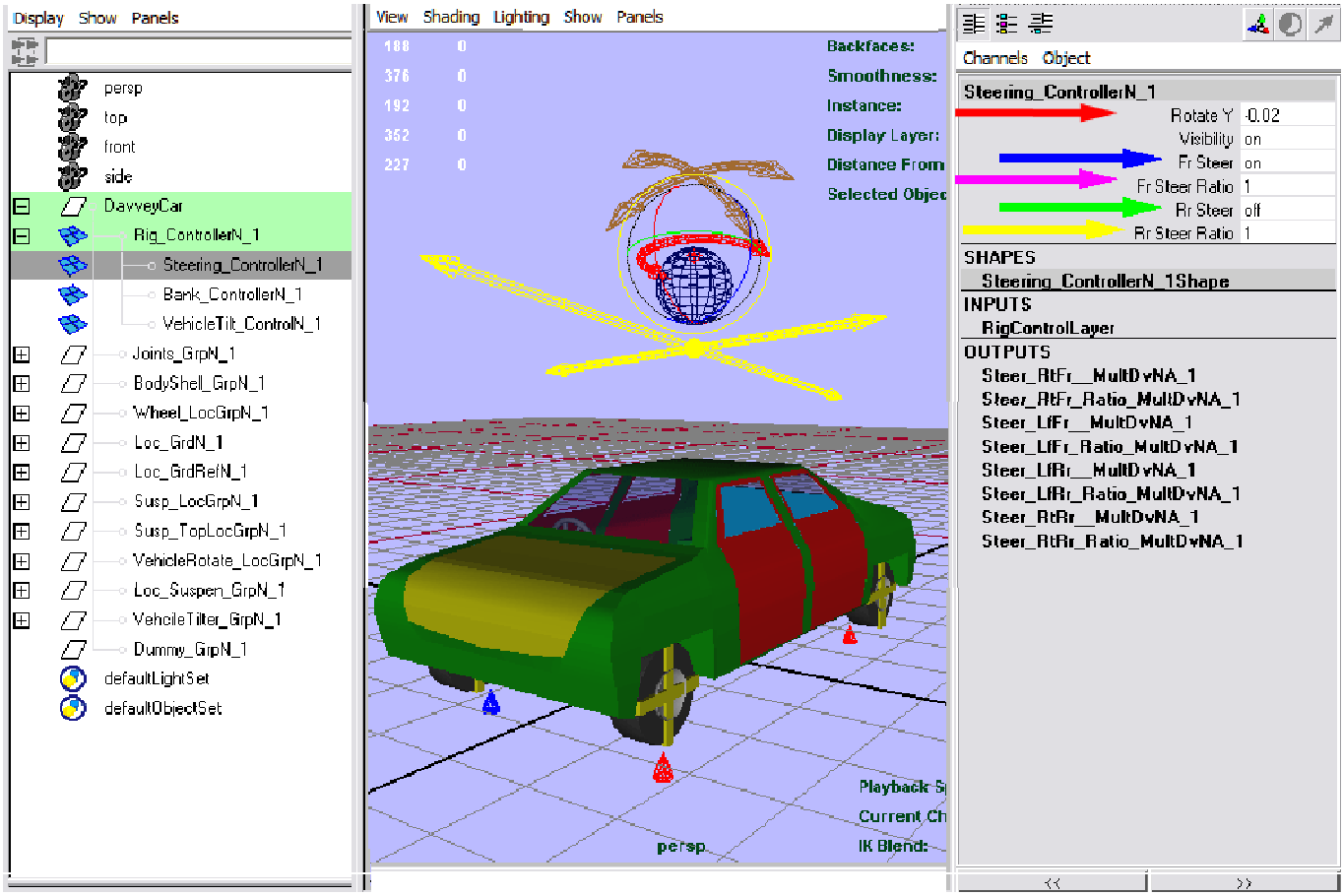
Structure of initial Vehicle is as follows:-

```
Top Node
 |___
   |
   |___BodyShell Group
       |
       |_____BodyWork_GrpN_1
           |     |
           |     |__Door_GrpN_1
           |     |      |
           |     |      |__LfFr_Door_GrpN_1
           |     |      |__RtFr_Door_GrpN_1
           |     |      |__LfRr_Door_GrpN_1
           |     |      |__RtRr_Door_GrpN_1
           |     |
           |     |__Fr_Hood_GrpN_1
           |     |__Rr_Trunk_GrpN_1 or Rr_Tailgate_GrpN_1
           |     |
           |     |                    or     (Rr_TopTailgate_GrpN_1
           |     |                            Rr_BotTailgate_GrpN_1)
           |     |
           |     |                           or (LfRr_Tailgate_GrpN_1
           |     |                           RtRr_TailDoor_GrpN_1)
           |     |
           |     |__Remaining Chassis parts(contains all geometry except Wheels.
           |
           |___Wheels Grp_1
               |___
               |Fr_Wheels_Grp_1
               |      |
               |      |__ LfFr_Wheel_Grp_1      (Wheel Group Naming must follow this structure
               |      |__ RtFr_Wheel_Grp_1        i.e LfFr_Wheel_Grp_1,  LfFr_Wheel _Grp_2
               |      |__ LfFr_Wheel_Grp_2        order is not important within each of the
               |      |__ RtFr_Wheel_Grp_2        Group Nodes.)
               |
               |Rr_Wheels_Grp_1
                      |
                      |__ LfRr_Wheel_Grp_1
                      |__ RtRr_Wheel_Grp_1
                      |__ LfRr_Wheel_Grp_2
                      |__ RtRr_Wheel_Grp_2
```

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Steering_ControllerN_1.



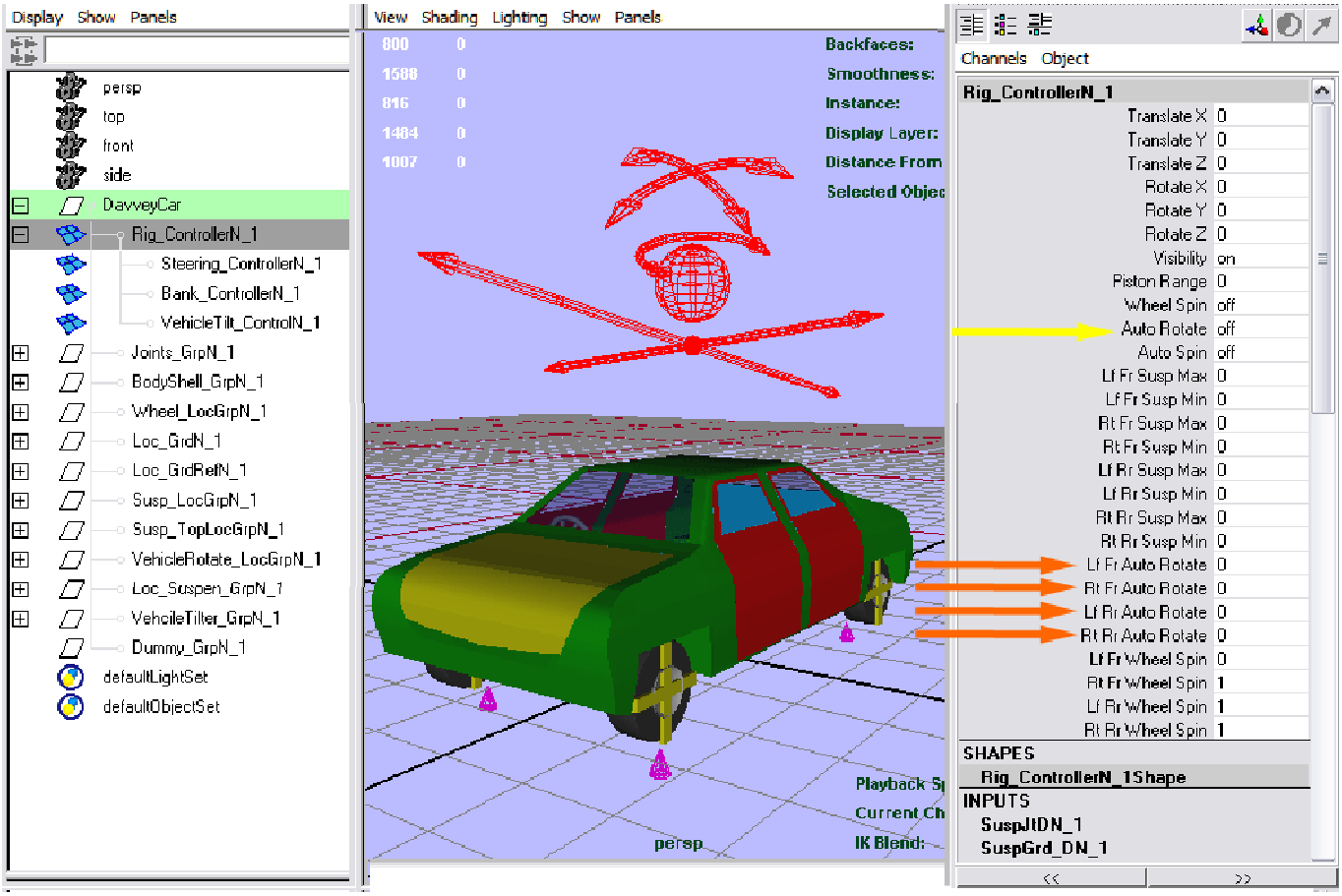
Controls steering for Front and Rear Wheels. The front steering is active by default, to activate Rear steering set Rr_Steer to On.

Setting the Fr_Steer Ratio and Rr_Steer Ratios to a negative value will reverse direction of rotation. Set Fr_Steer to Off will set vehicle to have rear steering. The ratio can be set anywhere between -1 and 1, this will reduce effect of rotation to simulate the Rear wheel steer you now see on some high performance cars.

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Wheel Rotation (Auto Rotate).



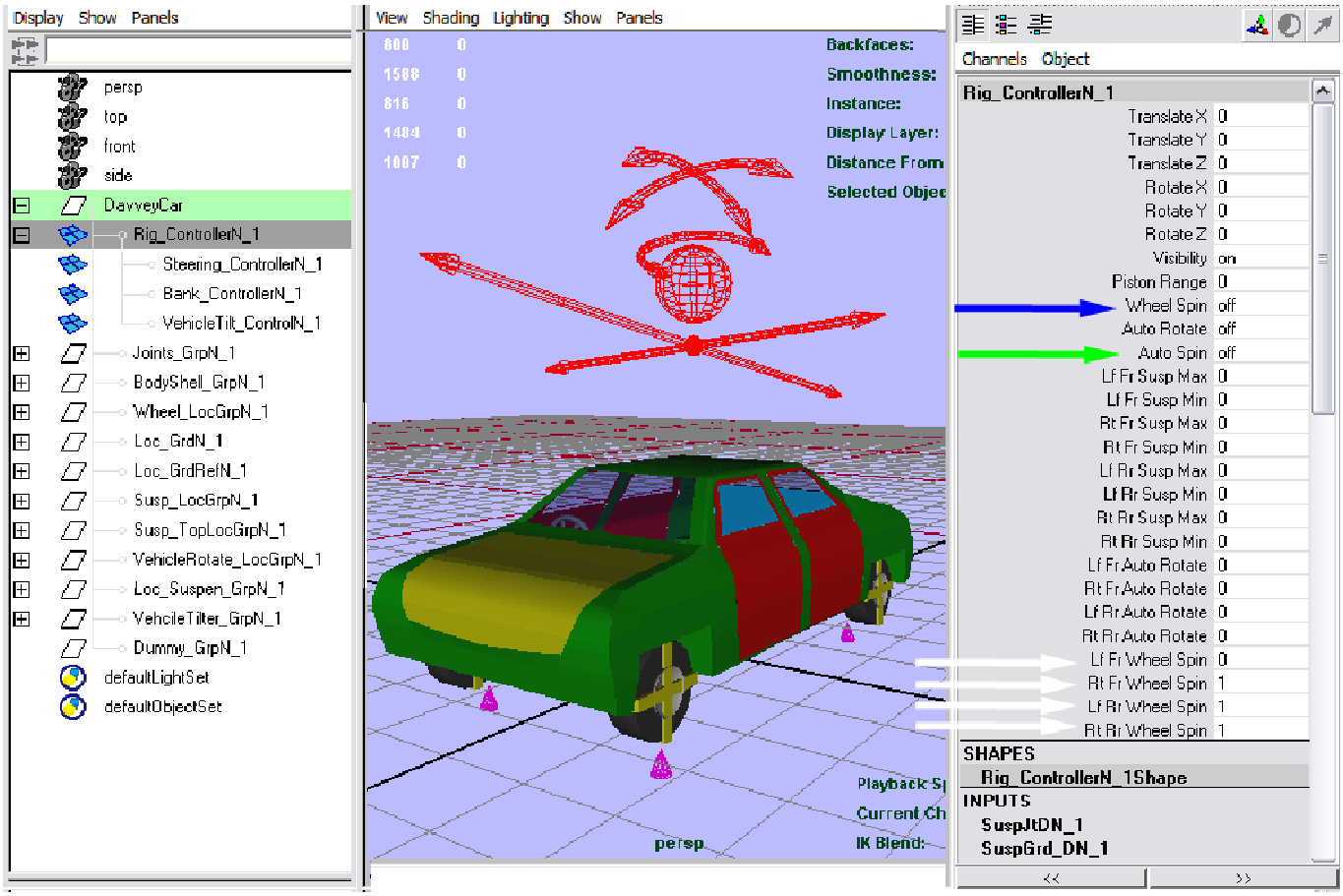
Auto Rotation of the Wheels is set by the AutoRotate attribute (Default On).

Once AutoRotate is set on each wheel rotation will be set to the calculated spin for travel in X and Y Axis. Each wheel can be individual switched of by setting LfFr_AutoRotate etc. to Zero. Also each wheel can be set to auto rotate at any speed greate or less than calculated value by setting LfFr_AutoRotate etc to a value between -10 and + 10 Again a value of zero will stop that wheel spinning. All fields will accept setting keys to control speed etc. The auto rotate will be automatically compensated when vehicle rotation in the Y axis occurs.

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Wheel Spin

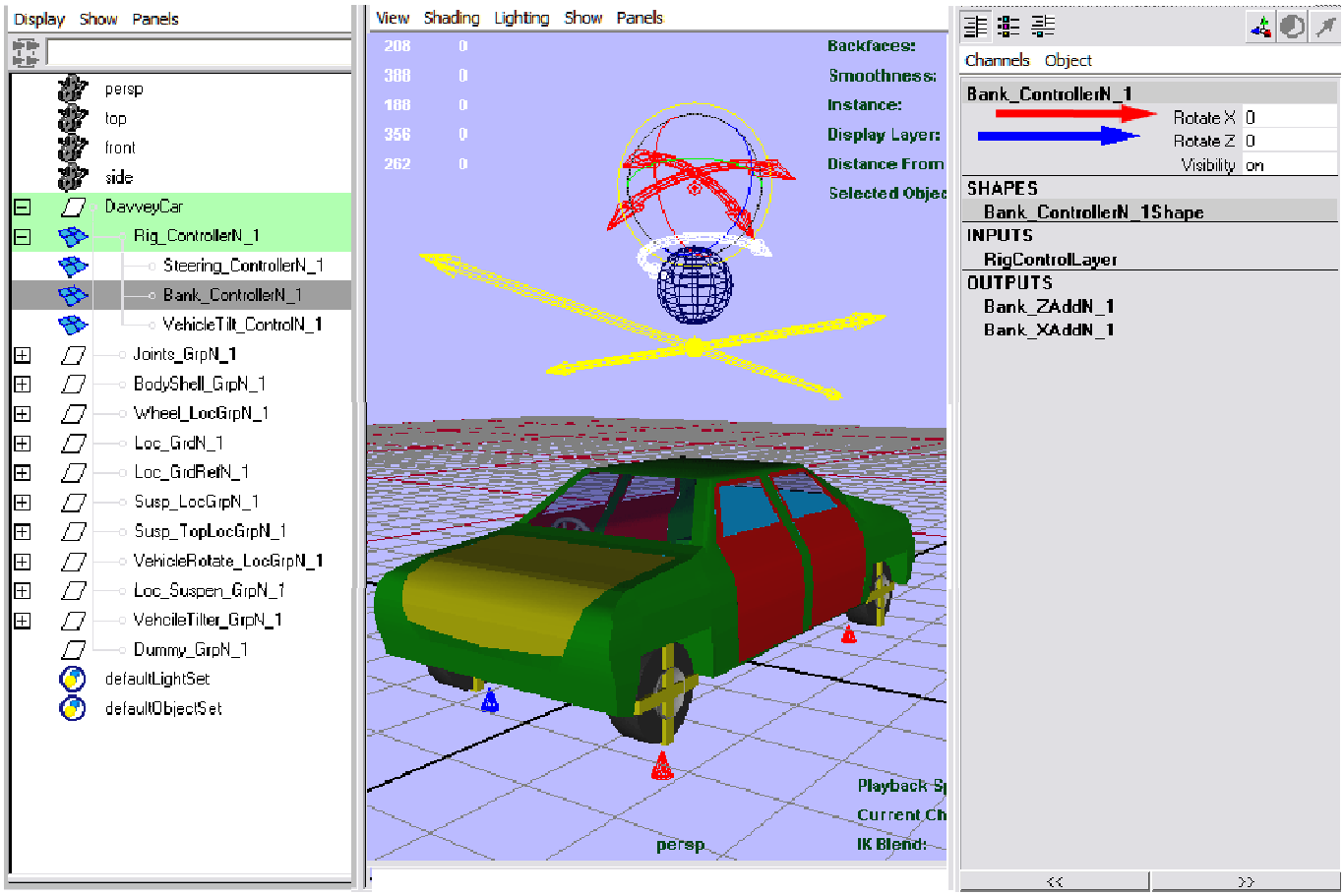


To activate Wheel Spin set AutoSpin to on. The speed of all four wheels can be varied by setting WheelSpin to any value between -10 and +10.

Once AutoSpin is set on and WheelSpin is set to any value except zero then each individual Wheel can be set to spin with any value between -10 and +10 . Again a value of zero will stop that wheel spinning. All fields can will accept setting keys and hitting play will activate Wheel Spin will be automatic. The Wheel Spin will be automatically compensated when vehicle rotation in the Y axis occurs.

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Bank_ControllerN_1



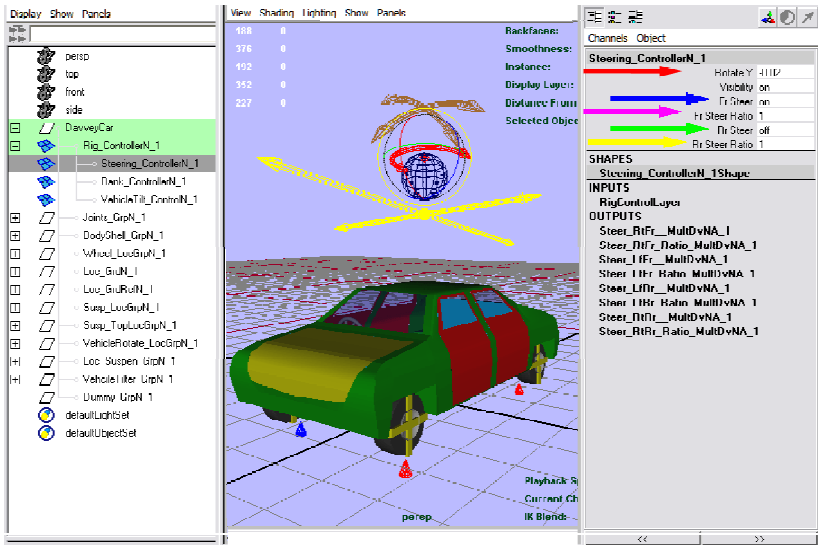
This will tilt the Chassis group as required. Also each individual Wheel can be moved up and down by moving each wheel Group Ground Locator to simulate ground variations.

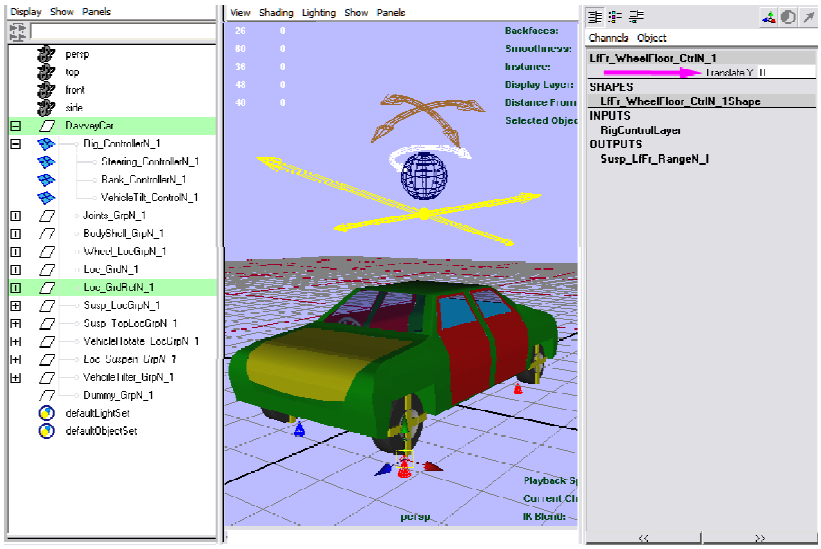
If these are the Geometry Constrained to a ground object or plane the movement will be automatic.

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Suspension SetUp.



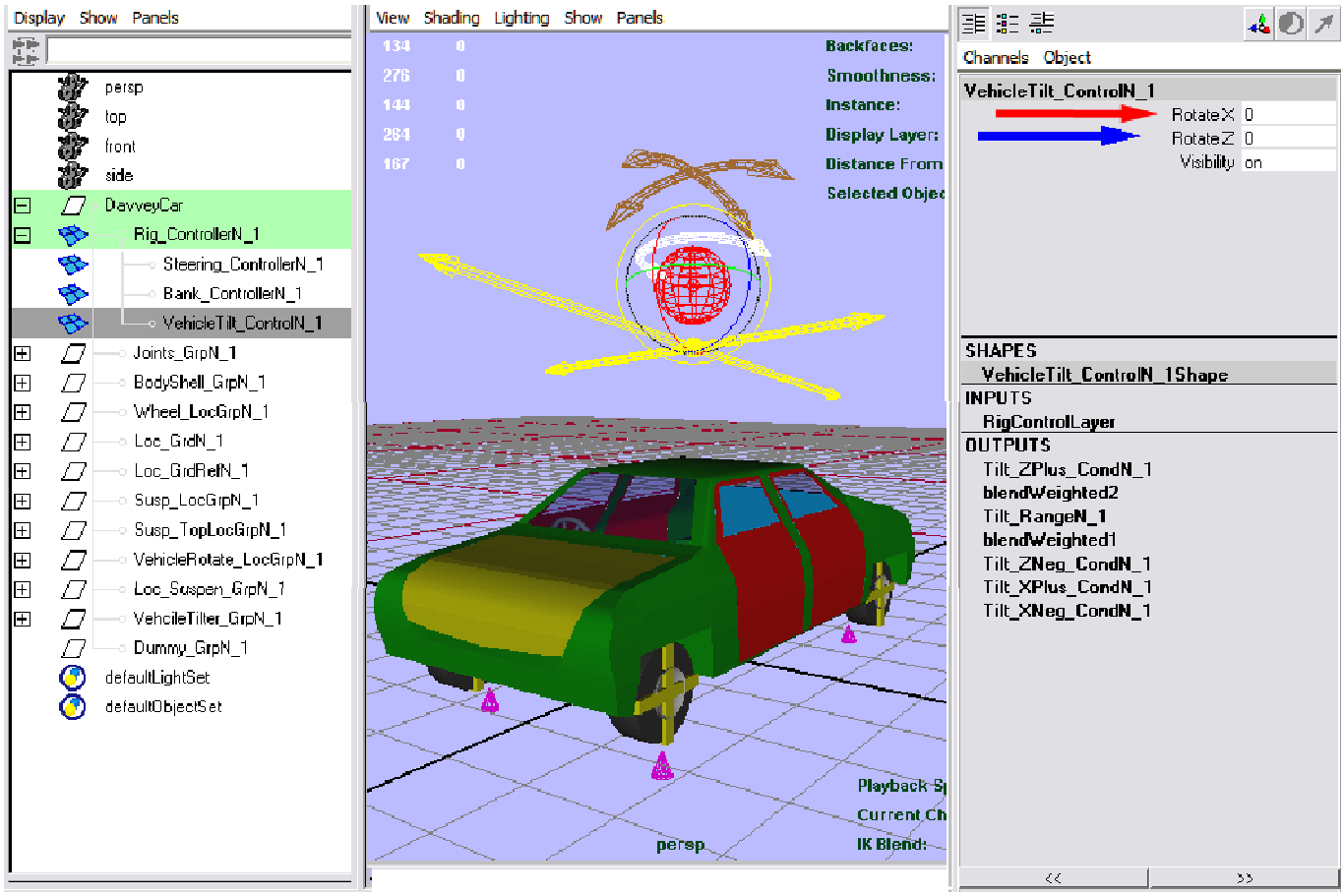


The maximum Suspension Travel is set using the attribute name Piston Range on the Main Rig Controller. This will set the maximum stroke for all four wheel groups. Each Wheel can have is own range of movement adjusted using the attributes named LfFr_Susp_Min LfFr_Susp_Max etc. Also each individual Wheel can be moved up and down by moving each wheel Group Ground Locator to simulate ground variations. If these are Geometry constrained to a ground object or plane the movement for each wheel will be automatic to the limit of the suspension Min Max travel settings.

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Tilt_ControllerN_1.



This will tilt the Whole Geometry of the Rig without tilting the rig controllers the Vehicle geometry will tilt +- 90 degrees in both X and Z Planes and still allow control of wheel travel etc.

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